

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

INTEGRATED AUDIO MODULE (IAM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Integrated Audio Module (IAM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Information and Entertainment System](#)

(415-00 Information and Entertainment System - General Information, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B119F-11	GPS Antenna - Circuit short to ground	NOTE: Circuit reference GPS_SIGNAL - Navigation antenna not connected to the integrated audio module - Navigation antenna circuit short circuit to ground	- Confirm that the navigation antenna is connected to the integrated audio module - Refer to the electrical circuit diagrams and check the navigation antenna circuit for short circuit to ground
B119F-13	GPS Antenna - Circuit open	NOTE: Circuit reference GPS_SIGNAL - Navigation antenna not connected to the integrated audio module - Navigation antenna circuit open circuit, high resistance	NOTE: This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis - Confirm that the navigation antenna is connected to the integrated audio module - Refer to the electrical circuit diagrams and check the navigation antenna circuit for open circuit, high resistance

B11A3-49	Gyroscope - Internal electronic failures	Integrated audio module internal gyroscope failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B121C-13	Hard Drive - Circuit open	Integrated audio module internal hard drive communication failure	NOTE: This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. Insert a CD and copy one track to the hard drive to confirm correct operation. Return vehicle to standard settings and delete the file. If the fault persists, install a new integrated audio module
B121C-49	Hard Drive - Internal electronic failures	Integrated audio module internal hard drive failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest

			level software. Insert a CD and copy one track to the hard drive to confirm correct operation. Return vehicle to standard settings and delete the file. If the fault persists, install a new integrated audio module
B1252-09	USB Port - Component failures	NOTE: Circuit reference USB_5B / USB_DATA_POS / USB_DATA_NEG / USB_DATA_GND / USB_SCR ▪ No USB function	▪ Refer to the electrical circuit diagrams and check the USB circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1252-19	USB Port - Circuit current above threshold	▪ Excessive current drawn by a universal serial bus device/cable ▪ Universal serial bus device internal failure ▪ Universal serial bus device cable internal failure	NOTE: If this DTC is logged, this suggests that the universal serial bus port has been overloaded, by a portable device or faulty cable ▪ Confirm universal serial bus port operation, and using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index ▪ Disconnect all the universal serial bus

			connected devices. Close all the doors and lock the vehicle, wait for a few minutes (to shut down the media orientated system transport ring). Restart the vehicle and check if the universal serial bus operation has recovered Using the manufacturer approved diagnostic system, clear the DTCs and retest. Advise the customer of the use of faulty devices/cables. Some devices will draw more current when charging such as iPads and can cause this DTC to log. This is not a permanent fault
B1296-4A	Navigation Map Data - Incorrect component installed	Navigation map does not match the market of the vehicle	NOTE: This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis Update the map data as necessary. Using the manufacturer approved diagnostic system, re-configure the integrated audio module with the latest level software. Using the manufacturer approved diagnostic system, check and update the car

			configuration file as necessary. Clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1A56-11	Antenna - Circuit short to ground	NOTE: Circuit reference AM_FM AM/FM antenna amplifier circuit short circuit to ground	Refer to the electrical circuit diagrams and check the AM/FM antenna amplifier circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1A56-13	Antenna - Circuit open	NOTE: Circuit reference AM_FM AM/FM antenna amplifier circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the AM/FM antenna amplifier circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D50-07	Digital Disk Player - Mechanical failures	Integrated audio module internal failure	NOTE: This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis

			Perform basic visual inspection for foreign matter inside disk player. Remove foreign matter if possible. Using the manufacturer approved diagnostic system, clear the DTCs and retest with multiple disks. If the fault persists, install a new integrated audio module
B1D55-11	Antenna#2 - Circuit short to ground	NOTE: Circuit reference FM2 FM/TMC antenna amplifier circuit short circuit to ground	Refer to the electrical circuit diagrams and check the FM/TMC antenna amplifier circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D55-13	Antenna#2 - Circuit open	NOTE: Circuit reference FM2 FM/TMC antenna amplifier circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the FM/TMC antenna amplifier circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D57-11	Antenna#4 Circuit - Circuit short to ground	VICS antenna circuit short circuit to ground	

			NOTE: Vehicle Information and Communication System (VICS) is a type of Traffic Message Channel system used in the Japan market only Refer to the electrical circuit diagrams and check the VICS antenna circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D57-13	Antenna#4 Circuit - Circuit open	VICS antenna circuit open circuit, high resistance	NOTE: Vehicle Information and Communication System (VICS) is a type of Traffic Message Channel system used in the Japan market only Refer to the electrical circuit diagrams and check the VICS antenna circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module

B1D78-11	Auxiliary Input - Circuit short to ground	NOTE: Circuit reference AUX1 L+ / AUX1 R+ / AUX1 LR- Auxiliary input 1 circuit short circuit to ground	Refer to the electrical circuit diagrams and check the auxiliary input 1 circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D78-13	Auxiliary Input - Circuit open	NOTE: Circuit reference AUX1 L+ / AUX1 R+ / AUX1 LR- Auxiliary input 1 circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the auxiliary input 1 circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D79-11	Microphone Input - Circuit short to ground	NOTE: Circuit reference MIC_1_POS / MIC_1_NEG Microphone 1 circuit short circuit to ground	Refer to the electrical circuit diagrams and check the microphone 1 circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
B1D79-13	Microphone Input - Circuit open	NOTE: Circuit reference MIC_1_POS / MIC_1_NEG Microphone 1 circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the microphone 1 circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module

P150E-00	Electronic Control Module Cooling Fan Circuit - No sub type information	Integrated audio module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
U2005-62	Vehicle Speed - Signal compare failure	- Anti-lock brake system control module power or ground circuit open circuit, high resistance - High speed CAN bus (chassis or powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - MOST network failure - Speed signal broadcast is not received by the integrated audio module - Anti-lock brake system control module internal failure	NOTES: - This DTC may be set when the vehicle is being transported - This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis - Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis and powertrain) circuits for short circuit to ground,

			short circuit to power, open circuit, high resistance - Refer to the relevant section of the workshop manual and test the MOST network - Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index. Clear the DTCs and retest
U200D-11	Control Module Output Power A - Circuit short to ground	NOTE: Circuit reference ANTENNA_AMP AM/FM antenna amplifier circuit short circuit to ground	Refer to the electrical circuit diagrams and check the AM/FM antenna amplifier circuit for short circuit to ground. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
U200D-13	Control Module Output Power A - Circuit open	NOTE: Circuit reference ANTENNA_AMP AM/FM antenna amplifier circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the AM/FM antenna amplifier circuit for open circuit, high resistance. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new integrated audio module
U2100-00	Initial Configuration Not Complete - No sub type information	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium

		▪ Touch screen internal failure ▪ MOST network failure ▪ Integrated audio module is not configured correctly	speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index ▪ Refer to the relevant section of the workshop manual and test the MOST network ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2101-00	Control Module Configuration Incompatible - No sub type information	▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Touch screen internal failure ▪ MOST network failure ▪ Integrated audio module is not configured correctly	NOTES: ▪ Only applicable to the Japanese market ▪ This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis ▪ Using the manufacturer approved diagnostic

			system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index ▪ Refer to the relevant section of the workshop manual and test the MOST network ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2200-41	Control Module Configuration Memory Corrupt - General checksum failure	▪ Integrated audio module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. If the fault persists, install a new integrated audio module
U2200-44	Control Module Configuration Memory Corrupt - Data memory failure	▪ Integrated audio module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. If the fault persists, install a new integrated audio

			module
U3000-41	Control Module - General checksum failure	Integrated audio module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. If the fault persists, install a new integrated audio module
U3000-45	Control Module - Program memory failure	Integrated audio module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. If the fault persists, install a new integrated audio module
U3000-46	Control Module - Calibration/parameter memory failure	Integrated audio module internal failure	NOTE: This DTC may be set even though no fault condition is present and should be ignored unless the customer has reported a navigation system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio

			module with the latest level software. If the fault persists, install a new integrated audio module
U3000-49	Control Module - Internal electronic failure	Integrated audio module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, re-configure the integrated audio module with the latest level software. If the fault persists, install a new integrated audio module
U3000-4A	Control Module - Incorrect component installed	- Incorrect hardware or software detected - Integrated audio module internal failure	- Check that the installed hardware is correct - Using the manufacturer approved diagnostic system, re-configure the integrated audio module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new integrated audio module
U3000-98	Control Module - Component or system over temperature	- Ventilation ducts obstructed - Integrated audio module internal failure	NOTE: Allow vehicle to cool before performing any diagnostic steps. Move vehicle into shade and operate climate control on a cool setting - Check the ventilation ducts for obstructions - Using the manufacturer approved diagnostic system, clear the DTCs

			and retest. If the fault persists, install a new integrated audio module
U3003-62	Battery Voltage - Signal compare failure	▪ Mismatch between the voltage at the integrated audio module and the voltage value broadcast on the CAN bus	▪ Refer to the electrical circuit diagrams and check the integrated audio module power and ground circuits for open circuit, high resistance